
Global Context and Feasibility Ablations for PIGNN

GBnetwork and case300

PIGNN–Attn–LS–PPC experimental report

1 Executive summary

This report consolidates the controlled global-context experiments for **case300** and **GBnetwork**. The experiments retain the PIGNN backbone and $K = 40$ recurrent corrections and change only the graph-context sidecar. G0 is the signed-log PIGNN reference; G1–G3 are global-context ablations; G4–G5 add dynamic sigmoid gating; and RANGE uses persistent master tokens. GraphKit and GridSFM values are included as external reference rows from the ppNR-v2 four-model report.

The main conclusions are:

- On **GBnetwork**, G2 (attention before local processing) gives the best voltage result among G1–G5, while G3 (attention after local processing) gives the best P/Q residuals among those variants.
- G4 and G5 do not improve consistently over their ungated counterparts; their sigmoid gates remain mostly open and therefore do not provide a useful selective safeguard.
- RANGE is now complete and test-scored. $M = 1$ is the best RANGE setting but lands inside the G1–G5 band rather than beyond it, and $M = 4$ and $M = 8$ are clearly worse: their masters collapse to identical representations (attention and embedding cosine +1.000). Persistent master tokens are therefore not a productive direction for GBnetwork.
- A per-bus residual breakdown shows PIGNN is *not* broadly worse than GridSFM on GBnetwork: G3’s median residual is lower on both channels. The entire gap is an extreme tail concentrated on four 22-kV buses (0.18% of the grid), which carry up to 20% of the total $|\Delta Q|$ mass, and it scales with $|Y_{ii}|$. GridSFM exceeds 10 pu in no bin of any breakdown, which reads as an output bound rather than better typical accuracy.
- Measured against a predictor that ignores the scenario entirely — each bus held at its own training-split mean — every model here loses on $|V|$ RMSE and on both median residuals, and wins only on bus angle. Two thirds of that angle win is gauge drift rather than flow accuracy (Section 8).
- The negative voltage R_σ^2 throughout this report is a *calibration* failure, not collapse: after each bus’s static profile is removed the models still explain 59–75% of the variance. Subtracting a per-bus offset fitted on the training split — no retraining — cuts $|V|$ RMSE by 2.0–2.7 $\times$, turns every R_σ^2 positive, and improves median $|\Delta Q|$ by 3–4 $\times$. It does not improve the worst-bus tail.
- Replacing the log-cosh physics term with MSE (G3-mse) is worse on every metric and an order of magnitude worse on the worst bus ($\max |\Delta P|$ 18.6 $\rightarrow$ 197.8 pu), which is the opposite of what a quadratic penalty was expected to do to the tail.

- The GridSFM values 1.10 and 4.81, like the corresponding PIGNN log fields, are *means of per-scenario maxima*, not global maxima over the complete test set. They are comparable as reducers when split, masks and precision also match.

Unless a table says otherwise, R^2 in this report means R_σ^2 of Section 7: $1 - (\text{RMSE}/\sigma)^2$ against a per-bus constant predictor. It is not a tracking measure; the tracking coefficient R_{bus}^2 appears there.

2 Experimental protocol

All G1–G5 and RANGE runs use the same PPC adapter, DC initialization, train/validation/test split, seed 42 and complex128 parquet decoding. The canonical settings are $d = 4$, $d_{hi} = 24$, eight heads, eight local attention layers, $K = 40$, voltage limits, direct updates, geometric-safe Armijo, `mse_weight=5`, graph-normalized log-cosh physics, signed-log residual features, learning rate 10^{-5} and cosine annealing. G1–G5 use the 37,022-row `GBnetwork` parquet. The contemporaneous `case300` G1–G5 runs use the 27,994-row comparison parquet; the separate 37,930-row authoritative v2 G0 is not a paired baseline for those runs.

The external GraphKit/GridSFM rows below come from a different campaign: one model per grid, seed 42, approximately 40 epochs, and the report’s original PPC-v2 data. They are useful architectural references, but are not a strict same-checkpoint comparison with the later G1–G5 campaign.

3 GBnetwork results

Setting	$ V $ RMSE	θ RMSE (deg)	mean case-max $ \Delta P $	mean case-max $ \Delta Q $
G0 signed-log reference	≈ 0.0302	≈ 0.889	not paired	not paired
G1 mean/max pre	0.030015	1.1963	5.3509	108.51
G2 attention pre	0.027269	0.9513	7.4611	68.119
G3 attention post	0.029390	1.0849	3.6316	46.803
G4 gated pre	0.030777	1.1578	7.1467	32.962
G5 gated post	0.031904	1.3042	4.4431	62.240
G3 with MSE physics	0.035169	1.4251	7.9892	205.45
GraphKit reference	0.0354	4.94	181	55.8
GridSFM reference	0.0319	2.14	1.10	4.81

Table 1: GBnetwork voltage metrics and mean per-scenario maxima. Every residual entry in the last two columns uses $N_s^{-1} \sum_s \max_i |\Delta P_{s,i}|$ (and analogously for Q); none is a global test-set maximum. External rows remain cross-campaign references. “G3 with MSE physics” is G3 with `physics_loss_form=mse` and nothing else changed; it is a paired ablation of G3, not a sixth global-context variant.

Setting	R_σ^2 on $ V $	R_σ^2 on θ
G1 mean/max pre	-0.611	0.971
G2 attention pre	-0.329	0.982
G3 attention post	-0.544	0.976
G4 gated pre	-0.694	0.973
G5 gated post	-0.820	0.965
G3 with MSE physics	-1.211	0.959
GraphKit reference	-1.241	0.502
GridSFM reference	-0.820	0.906

Table 2: GBnetwork coefficient of determination. G1–G5 use the report’s reference spreads, $\sigma_{|V|} = 0.02365$ pu and $\sigma_\theta = 7.0^\circ$, through $R_\sigma^2 = 1 - (\text{RMSE}/\sigma)^2$. GraphKit and GridSFM are the values printed in the four-model report.

Setting	mn P	mn Q	rms P	rms Q	p95 P	p95 Q	md P	md Q	$\leq 1e-2$ P	$\leq 1e-3$ P
G1	0.1173	0.2953	0.3024	2.6399	0.4355	0.7875	0.0455	0.1333	19.94%	3.66%
G2	0.1418	0.3056	0.3910	1.8777	0.5961	0.9023	0.0440	0.1336	20.87%	3.90%
G3	0.1094	0.2715	0.2477	1.3677	0.4332	0.7913	0.0411	0.1256	22.11%	4.29%
G4	0.1225	0.2897	0.4447	1.0635	0.4388	0.9575	0.0478	0.1272	20.09%	3.67%
G5	0.1114	0.2865	0.3246	1.5413	0.4069	0.8830	0.0468	0.1315	19.78%	3.66%
G3-mse	0.1252	0.3528	0.5837	4.8016	0.4754	0.8244	0.0452	0.1278	20.34%	3.85%
RANGE $M=1$	0.1095	0.2859	0.2695	2.7854	0.4104	0.7655	0.0465	0.1291	18.77%	3.43%
RANGE $M=4$	0.2771	0.5772	1.0370	5.0045	1.1322	1.7964	0.0896	0.2015	12.99%	2.74%
RANGE $M=8$	0.1696	0.6017	0.5973	4.8743	0.6326	2.0083	0.0649	0.1918	16.46%	4.23%
GraphKit	2.05	0.771	—	—	6.85	2.74	0.458	0.191	—	0.00%
GridSFM	0.0828	0.266	—	—	0.304	0.913	0.0432	0.156	—	0.03%

Table 3: GBnetwork residual distributions, exact-complex test summary fields, in pu on a 100-MVA base. **Correction to the previous version of this table:** the columns previously headed “mean P” and “mean Q” carried the driver’s RMSE $\Delta P/\Delta Q$ fields, not the pooled means, and the columns headed “tol” were the $\leq 10^{-3}$ pu fractions and not the 10^{-6} pu tolerance. Both are now separated and relabelled; no underlying run changed. The 10^{-6} pu fraction is $\leq 0.02\%$ for every PIGNN row here and is omitted. The external GraphKit/GridSFM rows are the four-model report’s printed mean/median/p95 fields; that report did not print RMSE or the 10^{-2} pu fraction, so those cells are blank, and its “tol” column is used in the $\leq 10^{-3}$ position.

The PIGNN G2 voltage RMSE is about 9.7% below the historical G0 value. G3 reduces the PIGNN mean case-max residuals relative to G2, especially reactive power, but does not simultaneously minimize voltage error.

4 case300 results

Setting	$ V $ RMSE	θ RMSE (deg)	mean case-max $ \Delta P $	mean case-max $ \Delta Q $
G1 mean/max pre	0.032798	4.9682	10.914	10.885
G2 attention pre	0.027823	4.5222	5.463	7.260
G3 attention post	0.026900	4.5463	6.369	9.050
G4 gated pre	0.032155	5.4597	10.160	11.566
G5 gated post	0.034005	5.3795	8.932	11.859
GraphKit reference	0.0237	4.76	10.4	6.82
GridSFM reference	0.0247	27.7	1.22	1.23

Table 4: case300 comparison. The G1–G5 rows use the 27,994-row campaign; the external rows use the four-model report’s case300 campaign.

Global context does not provide a uniform case300 improvement. G3 obtains the lowest magnitude RMSE, while G2 obtains the lowest reported mean case-max P/Q residuals. The angle errors remain large for all five variants. On GBnetwork the per-bus/per-edge split in Section 8 shows roughly a third to two thirds of such an error to be gauge drift rather than flow error; the same split has not been measured on case300, so the case300 angle column should be read as an upper bound on the flow-relevant error rather than as its size.

5 RANGE final test results

The three RANGE training jobs were stopped by the four-hour Slurm wall clock (state TIMEOUT) after 38, 33 and 33 of the planned 40 epochs for $M = 1, 4, 8$; their best checkpoints were written at epochs 35, 27 and 33 respectively. Because the in-job test block never ran, the saved checkpoints were re-evaluated in a separate test-only array (job 797689, three tasks, all COMPLETED) on the same 12,344-scenario test split, seed 42, with the identical

flags. The values below are therefore final test values, not validation values, but they belong to slightly under-trained models.

Masters	best ep.	$ V $ RMSE	θ RMSE	$R^2_{\sigma, V }$	$R^2_{\sigma,\theta}$	mean c-max $ \Delta P $	mean c-max $ \Delta Q $
$M = 1$	35/38	0.03253	1.016°	-0.891	+0.979	4.007	100.56
$M = 4$	27/33	0.04455	2.966°	-2.548	+0.821	31.426	211.25
$M = 8$	33/33	0.04683	1.604°	-2.921	+0.947	17.843	205.35

Table 5: RANGE final test metrics. “best ep.” is the epoch of the loaded checkpoint over the epochs actually reached before the wall clock. The R^2_{σ} values are recomputed here with the same reference spreads used for G1–G5, $\sigma_{|V|} = 0.02365$ pu and $\sigma_{\theta} = 7.0^\circ$, so that this table can be read against Table 2. **These are not the driver’s printed R^2 fields**, which for these three runs read +0.610/+0.269/+0.193 on $|V|$. The two differ only in the baseline predictor in the denominator, and the gap is large. `train_valid_test.py` pools every bus-scenario value and subtracts a single global mean, so its baseline is “predict one number everywhere”; measured on this parquet that spread is $\sigma_{|V|}^{\text{global}} = 5.23 \times 10^{-2}$ pu, because it also contains the *static* bus-to-bus differences in nominal voltage level. The spread used throughout this report subtracts each bus’s own mean first, so its baseline is “predict each bus’s usual voltage”, giving $\sigma_{|V|}^{\text{per-bus}} = 2.38 \times 10^{-2}$ pu. Only the latter measures what the surrogate must actually learn — how voltages move as the loading changes — since per-bus nominal levels are already available from `vn_kv` and the admittance matrix. The GraphKit and GridSFM drivers emit no R^2 at all, so every external R^2 in this report is likewise a per-bus-baseline hand computation of R^2_{σ} ; mixing the two conventions in one column would make a model with a *worse* $|V|$ RMSE appear to be the only one with a positive value.

Masters	agg./bcast. entropy	eff. masters	attn. cosine	emb. cosine	self-loop mass
$M = 1$	1.000 / 1.000	1.000	+0.000	+0.000	50.0%
$M = 4$	1.000 / 1.000	4.000	+1.000	+1.000	20.0%
$M = 8$	1.000 / 1.000	8.000	+1.000	+1.000	11.2%

Table 6: RANGE master diagnostics on the test split. The effective master count equals the nominal count only because attention is uniform; for $M = 4$ and $M = 8$ the master-attention and master-embedding cosine similarities are both +1.000, so the masters are duplicates rather than specialists. Token utilization is exactly uniform ($1/M$ each) in both cases.

The final test values confirm the interim reading and sharpen it. $M = 1$ is the best RANGE setting on every metric except pooled RMSE ΔQ , and its voltage RMSE (0.0325) and angle RMSE (1.02°) place it between G1 (0.0300, 1.20°) and G5 (0.0319, 1.30°) — that is, *inside* the G1–G5 band rather than beyond it. Its pooled mean and median residuals are likewise indistinguishable from G3’s (mean $|\Delta P|$ 0.1095 vs. 0.1094). Adding masters is strictly harmful: $M = 4$ roughly doubles every pooled residual statistic and triples the angle error, and $M = 8$ recovers only part of that.

The collapse diagnostics explain why. With $M > 1$ the masters converge to a single shared representation, so the extra tokens add parameters and dilute the per-token attention mass (self-loop mass falls from 50% to 11%) without adding any distinct global channel. A single master avoids the collapse trivially, but then the sidecar reduces to the same one-token global summary already tested as G2/G3 — and it performs like it. On this evidence, persistent master tokens are not a productive direction for GBnetwork, and the $M = 1$ result should not be read as a positive RANGE finding.

One caveat: all three runs stopped 2–7 epochs short and $M = 1$ was still improving at the wall clock (validation loss 1.60×10^{-2} at epoch 35 against 1.69×10^{-2} at 38, without a clear plateau). A rerun with a longer wall clock could move these numbers, but not by the factor that would be needed to separate $M = 1$ from G2/G3.

6 Where the residual mass actually is

Every table above reduces the residual to a scalar, which cannot distinguish a model that is uniformly slightly worse from one that is equal in the bulk and catastrophic on a handful of buses. A dedicated diagnostic (alex array 4123954/4124031, six tasks, all **COMPLETED**) re-evaluated the G1–G5 and GridSFM v2 checkpoints on the identical 12,344-scenario test split, identical P/Q masks and a final complex128 recomputation, and broke the residuals down by bus type, nominal voltage, $|Y_{ii}|$, degree and transformer adjacency.

The rescore reproduces the published voltage metrics to within 0.8% on all six models ($|V|$ within 0.12%, worst angle deviation -0.77% on G3), so the breakdowns below describe the same checkpoints as Tables 1–3 and not a different evaluation.

Setting	$ \Delta P $				$ \Delta Q $			
	median	RMSE	glob. max	%>1 pu	median	RMSE	glob. max	%>10 pu
G1	0.0455	0.302	36.1	1.03%	0.1333	2.642	172.4	0.057%
G2	0.0442	0.398	37.4	2.11%	0.1338	1.878	157.2	0.074%
G3	0.0410	0.244	18.6	1.00%	0.1255	1.354	141.4	0.061%
G4	0.0478	0.465	304.2	1.19%	0.1273	1.069	230.2	0.062%
G5	0.0468	0.324	58.8	0.79%	0.1315	1.539	141.9	0.061%
GridSFM	0.0432	0.141	2.4	0.05%	0.1557	0.458	9.9	0.000%

Table 7: Bulk versus tail on GBnetwork, pu on a 100-MVA base. “glob. max” is the true $\max_{s,i} |\Delta \cdot_{s,i}|$ over all bus–scenario entries, a quantity none of the earlier tables reported. **PIGNN is not broadly worse than GridSFM**: G3’s median residual is lower on both channels (0.0410 vs. 0.0432 for P, 0.1255 vs. 0.1557 for Q) and its pooled mean $|\Delta Q|$ is within 2% of GridSFM’s (0.271 vs. 0.266, Table 3). The whole gap is in the extreme tail: RMSE differs by 1.7–3.3 $\times$ and the global maximum by one to two orders of magnitude.

The tail is four buses

Splitting by nominal voltage localizes it almost completely. The 22-kV bin contains 49,376 entries = 4 buses $\times$ 12,344 scenarios — four buses out of 2224, 0.18% of the grid.

Setting	mean $ \Delta Q $ at 22 kV	%>10 pu there	glob. max there	share of total Q mass
G1	27.71	24.8%	172.4	20.3%
G2	17.23	23.4%	157.2	12.2%
G3	11.80	22.3%	141.4	9.4%
G4	8.55	25.0%	230.2	6.4%
G5	16.02	24.8%	141.9	12.1%
GridSFM	0.217	0.0%	3.5	0.2%

Table 8: The four 22-kV buses. “Share of total Q mass” is $\bar{q}_{22}n_{22}/(\bar{q}_{\text{pool}}n_{\text{pool}})$, the fraction of the grid’s entire pooled $|\Delta Q|$ contributed by these four buses. For G1 they carry a fifth of it. The same buses cost GridSFM 0.2%. The P channel shows the identical pattern more weakly (1.6–5.0% of P mass for G1–G5, 0.2% for GridSFM).

A paired G3 run with the physics loss changed from log-cosh to MSE (`physics_loss_form=mse`, every other flag identical) is queued as helma job 797773 to test whether a quadratic penalty suppresses this tail; a first attempt on an alex A40 died in epoch 1 with a CUDA out-of-memory at the attention logits, which is a GPU-capacity failure and not a property of the objective. Its result is not included here.

It tracks $|Y_{ii}|$, not the grid as a whole

$ Y_{ii} $ bin	entries	G3 mean $ \Delta Q $	G3 %>10 pu	GridSFM mean	GridSFM %>10
10^0 – 10^1	4,073,520	0.153	0.000%	0.142	0.000%
10^1 – 10^2	7,319,992	0.243	0.000%	0.205	0.000%
10^2 – 10^3	7,221,240	0.244	0.000%	0.351	0.000%
10^3 – 10^4	2,703,336	0.280	0.000%	0.322	0.000%
10^4 – 10^5	1,234,400	0.738	0.892%	0.355	0.000%
10^5 – 10^6	148,128	2.267	1.971%	0.953	0.000%

Table 9: $|\Delta Q|$ against the magnitude of the diagonal admittance. Below $|Y_{ii}| = 10^4$ PIGNN G3 matches or beats GridSFM and never exceeds 10 pu; above it, G3 degrades by 3–9 $\times$ while GridSFM does not. The degree breakdown agrees: the single degree-15 bus has G3 mean 6.87 pu with 8.4% of entries above 10 pu, against GridSFM’s 0.474 pu and 0.0%.

Two conclusions follow, and both revise the reading in the next section.

First, **the PIGNN deficit is a stiffness-localized tail, not a representation-capacity deficit**. The failure sits exactly on the worst-conditioned nodes, is confined to well under 1% of entries, and leaves the bulk distribution competitive or better. This is the failure mode a diagonal equilibration of the residual features is meant to address, and it is the clearest evidence so far that the conditioning hypothesis — rather than receptive field or parameter count — explains the large-grid gap.

Second, **GridSFM’s advantage looks like a bound, not better typical accuracy**. Its %>10 pu is exactly 0.000 in *every* bin of every breakdown — bus type, voltage level, $|Y_{ii}|$, degree and transformer adjacency alike — while its median $|\Delta Q|$ is the worst of the six models. A model that is uniformly capped and slightly loose in the middle is what a clipped magnitude head and a bounded angle residual around a DC prior would produce. The corresponding PIGNN mechanism, the geometric-safe Armijo line search, bounds the *step* rather than the *output*, which does not prevent a badly-conditioned bus from accumulating error over $K = 40$ iterations.

7 Notation: three different R^2 , and the calibration

Three distinct coefficients appear in the literature and in this project’s logs, and earlier revisions of this report used the word “ R^2 ” for more than one of them. Every table below names which one it is. Write v for the reference $|V|$ at a bus in a scenario, $\hat{v}$ for the prediction, $\bar{v}_i$ for bus i ’s mean over scenarios, and fit $\hat{v} = a v + b + \varepsilon$ by least squares.

Symbol	Definition	Null model it scores 0 against	Calibration
R^2_{pool}	Pearson R^2 of $\hat{v}$ on v , pooled over all buses and scenarios	one number for every bus and scenario	changes
R^2_{bus}	the same after subtracting $\bar{v}_i$ from $\hat{v}$ and v alike	each bus at its own mean	invariant
R^2_{σ}	$1 - \sum(\hat{v} - v)^2 / \sum(v - \bar{v}_i)^2$	each bus at its own mean	changes

R^2_{pool} is the diagnostic GridSFM’s section 6.2 defines, and it is the right one for a multi-grid foundation model, which must reproduce a static voltage profile it has never seen. Every campaign here trains one model per grid on that grid, where that profile is memorisable from the training split; a predictor that emits each bus’s training mean and ignores the scenario entirely scores $R^2_{\text{pool}} = 0.798$ on GBnetwork. So R^2_{bus} is the tracking question in this setting, and R^2_{pool} is kept only for comparability with the published numbers.

R^2_{σ} is what earlier revisions of this report printed. It is not a tracking measure: it charges the model for bias and for amplitude error as well. Writing $\sigma^2 = \frac{1}{N} \sum (v - \bar{v}_i)^2$ and σ_{ε}^2 for

the variance of ε , the mean square error splits exactly into three terms,

$$\text{RMSE}^2 = \underbrace{b^2}_{\text{bias}} + \underbrace{(a-1)^2\sigma^2}_{\text{amplitude}} + \underbrace{\sigma_\varepsilon^2}_{\text{noise}}, \quad \text{so} \quad R_\sigma^2 = 1 - \frac{b^2}{\sigma^2} - (a-1)^2 - \frac{\sigma_\varepsilon^2}{\sigma^2}.$$

R_{bus}^2 normalises the noise term by the variance of the *prediction* rather than of the reference, so it is not a simple difference away from R_σ^2 ; the two answer different questions and should not be converted into one another.

Three relations are used below, and each was checked against the measurements rather than assumed. (i) Calibration removes exactly the b^2/σ^2 term, so $R_{\sigma,\text{cal}}^2 = 1 - (a-1)^2 - \sigma_\varepsilon^2/\sigma^2$; over the 26 stabilization runs this held to 1.4×10^{-2} . (ii) R_{bus}^2 and the slope are unchanged by calibration, since a per-bus offset cannot alter a regression on per-bus deviations; measured deviation $< 10^{-6}$ on all 26. (iii) $R_{\sigma,\text{cal}}^2 \leq R_{\text{bus}}^2$, with the gap being the amplitude error; observed on all 26, from 0.0006 to 1.22.

Post-hoc per-bus calibration. The b^2/σ^2 term is removable without touching the model. For each bus, take the mean prediction over the *training* split, align it to the training reference mean, and subtract that fixed offset from every unseen test scenario:

$$\hat{v}_{i,s}^{\text{cal}} = \hat{v}_{i,s} - (\bar{\hat{v}}_i^{\text{train}} - \bar{v}_i^{\text{train}}),$$

and the circular equivalent for θ . No test data enters the fit, so this is a correction a deployed model could carry. Because it is fitted on one split and applied to another it does not transfer perfectly: across the 26 stabilization runs the realised improvement in R_σ^2 differs from the fitted b^2/σ^2 by a median of 1.2% and at most 8.4%. Tables that report it use the form *raw* $\rightarrow$ *calibrated*. R_{bus}^2 and the slope are mathematically unchanged by it — a per-bus offset cannot alter a regression on per-bus deviations — which is used below as a consistency check on the implementation.

8 What the metrics were measuring

Every ranking above rests on $|V|$ RMSE, θ RMSE and R^2 . Three follow-up runs on the same checkpoints, split and masks show that each of the three was answering a different question from the one it was being read as.

A predictor that ignores the scenario. The natural null model here is not $V = 1.0$ everywhere but “each bus sits at its own usual voltage”: take each bus’s mean $|V|$ and circular-mean θ over the *training* third and emit them regardless of loading. Scored on the same test third (alex job 4124582) it reproduces the reference spreads exactly — $|V|$ RMSE 2.3412×10^{-2} against a measured $\sigma_{|V|}^{\text{per-bus}} = 2.3412 \times 10^{-2}$ pu, θ RMSE 6.846° against 6.846° — which is the definitional check that it scores $R^2 = 0$ on both channels.

Setting	$ V $ RMSE	θ RMSE	med $ \Delta P $	med $ \Delta Q $	RMSE ΔQ	max $ \Delta Q $
per-bus constant	0.02341	6.846°	0.0033	0.0015	0.088	3.4
G1	0.03002	1.197°	0.0455	0.1332	2.642	172.3
G2	0.02727	0.946°	0.0441	0.1337	1.874	139.1
G3	0.02940	1.073°	0.0410	0.1255	1.355	141.4
G4	0.03078	1.184°	0.0478	0.1272	1.076	249.2
G5	0.03191	1.302°	0.0468	0.1314	1.539	141.9
GridSFM	0.03186	2.144°	0.0432	0.1557	0.458	9.9
G3-mse	0.03517	1.425°	0.0452	0.1278	4.802	215.6

Table 10: Against a model that has learned nothing about the scenario. It wins on $|V|$ RMSE, on both median residuals, on pooled RMSE ΔQ and on the worst bus in the corpus. The only column any trained model wins is the bus angle. Residuals in pu on a 100-MVA base.

The angle column does not mean what it appears to. Branch flows depend on $\theta_i - \theta_j$, never on the absolute bus angle, so a common per-scenario offset inflates per-bus θ RMSE without moving a single flow. Splitting the error accordingly (deg):

Setting	per-bus	of which gauge	de-gauged	per-edge $\Delta\theta_{ij}$
per-bus constant	6.846	4.029	5.532	1.000
G2	0.946	0.330	0.887	0.376
G3	1.073	0.483	0.958	0.403
G5	1.302	0.837	0.997	0.425
GridSFM	2.144	1.575	1.454	0.447

Table 11: Angle error RMSE. “gauge” is the circular mean of the bus-angle error within each scenario. On per-bus RMSE G2 beats the constant predictor by $7.2\times$ and GridSFM by $2.3\times$; on the quantity the flows see those margins fall to $2.7\times$ and $1.2\times$. More than two thirds of GridSFM’s angle error is gauge, which is the criticism its own paper levels at gridfm-graphkit. The angle ranking in Table 1 and Table 5 is therefore substantially a ranking of gauge drift.

Two different R^2 , and only one of them is about tracking. GridSFM’s section 6.2 diagnostic is a pooled regression of prediction against reference, reported with a slope; its R^2 is a Pearson R^2 and asks whether the model orders scenarios correctly. The R^2 in Tables 2 and 5 of this report is $1 - (\text{RMSE}/\sigma)^2$, which also charges the model for bias and for amplitude error. They coincide only when the slope is 1 and the bias is zero. Furthermore, pooling over buses is right for GridSFM — a multi-grid foundation model must predict a grid’s static profile because it has not seen it — but wrong here, where one model is trained per grid and that profile is memorisable from the training split.

Setting	pooled		per-bus		R_σ^2
	slope	R_{pool}^2	slope	R_{bus}^2	
per-bus constant	0.798	0.798	0.000	0.000	−0.000
G1	0.587	0.681	0.487	0.621	−0.644
G2	0.640	0.740	0.648	0.709	−0.357
G3	0.578	0.705	0.491	0.686	−0.577
G4	0.554	0.676	0.510	0.652	−0.728
G5	0.539	0.643	0.476	0.592	−0.857
GridSFM	0.575	0.631	0.688	0.752	−0.852
G3-mse	0.629	0.611	0.316	0.513	−1.258

Table 12: R^2 on $|V|$ under the three readings of Section 7. The first row is the control: a predictor that reproduces each bus’s mean and nothing else scores 0.798 pooled — four fifths of the pooled variance on this grid is the static profile, not the scenario. **Every model retains $R^2 = 0.59$ – 0.75 after that profile is removed**, so none of them is collapsed in the sense the report previously claimed; the negative SSE column is measuring something else.

What the negative column is measuring: a per-bus offset. Writing the prediction as $\hat{v} = a v + b + \varepsilon$ with ε uncorrelated with v ,

$$\text{RMSE}^2 = \underbrace{b^2}_{\text{bias}} + \underbrace{(a-1)^2\sigma^2}_{\text{amplitude}} + \underbrace{\sigma_\varepsilon^2}_{\text{noise}}, \quad \sigma_\varepsilon^2 = (1 - R^2)\sigma_v^2.$$

Setting	$ V $ RMSE	bias	amplitude	noise
per-bus constant	0.02341	0.00006	0.02341	0.00000
G2	0.02727	0.02410	0.00825	0.00972
G3	0.02940	0.02572	0.01191	0.00779
G5	0.03191	0.02796	0.01227	0.00924
GridSFM	0.03186	0.02960	0.00730	0.00926
G3-mse	0.03517	0.02961	0.01602	0.00981

Table 13: Decomposition of the $|V|$ error, pu. The control row validates it: for a constant predictor the bias vanishes, the amplitude term is the full σ and there is no noise, so $\text{RMSE} = \sigma$ exactly. For every trained model the *bias* term alone exceeds the entire scenario spread $\sigma = 0.02341$: each bus’s predicted mean voltage is systematically wrong by more than the whole quantity being predicted.

Removing the offset, and what it does not fix. This is a claim about a correction learnable without the test set, so it was fitted where the model’s own training data lives — a per-bus offset measured on the training third, subtracted at test time, nothing retrained (alex job 4124803).

Setting	$ V $			residual medians				
	RMSE	R_σ^2 (pred.)		$ \Delta P $	$ \Delta Q $	$\Delta\theta_{ij}$	max $ \Delta Q $	
G1	0.0300→ 0.0150	−0.644 → +0.592	+0.592	0.046→0.027	0.133→0.044	0.41→0.29	172→138	
G2	0.0273→ 0.0128	−0.357 → +0.703	+0.704	0.044→0.030	0.134→0.042	0.38→0.27	139→ 197	
G3	0.0294→ 0.0142	−0.577 → +0.630	+0.630	0.041→0.024	0.126→0.037	0.40→0.28	141→ 160	
G4	0.0308→ 0.0144	−0.728 → +0.621	+0.621	0.048→0.026	0.127→0.040	0.41→0.29	249→ 357	
G5	0.0319→ 0.0154	−0.857 → +0.569	+0.569	0.047→0.025	0.131→0.042	0.43→0.30	142→88	
GridSFM	0.0319→ 0.0118	−0.852 → +0.746	+0.747	0.043→0.014	0.156→0.039	0.45→0.23	9.9→9.7	
G3-mse	0.0352→ 0.0176	−1.258 → +0.438	—	0.045→0.019	0.128→0.031	0.40→0.28	216→56	

Table 14: Before and after subtracting a per-bus offset fitted on the training split. The “predicted” column is what the decomposition above says should happen; measurement agrees to within 0.001 on all six, which is the check that the decomposition is the right one. $|V|$ RMSE improves 2.0–2.7 $\times$, every model passes the constant predictor for the first time, median $|\Delta Q|$ improves 3.1–4.0 $\times$, and per-edge $\Delta\theta_{ij}$ improves about 30%. **The extreme tail does not follow**: three of the six report a *larger* worst-bus $|\Delta Q|$ afterwards.

Log-cosh against MSE in the physics term. Section 6 localised the PIGNN deficit to a handful of very large residuals, which makes a quadratic penalty the obvious thing to try: MSE weights a residual of 10 pu a hundred times more than one of 1 pu, where log-cosh weights it about ten times. G3 was therefore rerun with `physics_loss_form=mse` and every other flag unchanged (helma job 797773, 40/40 epochs; a first attempt on an alex A40 died in epoch 1 with a CUDA out-of-memory, which is a GPU-capacity failure and not a property of the objective).

It is worse on every axis, and worst exactly where it was meant to help. Voltage RMSE rises from 0.02940 to 0.03517, the angle from 1.073° to 1.425° , R_{bus}^2 falls from 0.686 to 0.513 and the slope from 0.491 to 0.316 — so the quadratic term costs tracking as well as accuracy. The worst bus goes the wrong way by an order of magnitude: $\max |\Delta P|$ $18.6 \rightarrow 197.8$ pu, $\max |\Delta Q|$ $141.4 \rightarrow 215.6$ pu. This is the failure mode the choice of log-cosh was guarding against: with a squared penalty the few stiff buses of Section 6 dominate the gradient and the fit is spent on them.

The one place it does better is under calibration, and revealingly so. G3-mse carries the largest *amplitude* error of any setting here (0.01602 pu against 0.00730–0.01227 for the rest) and an ordinary bias, and once the bias is removed its pooled ΔQ collapses — RMSE $4.802 \rightarrow 0.221$, a 22 $\times$ improvement, and $\max |\Delta Q|$ $215.6 \rightarrow 56.1$ — whereas calibrating G3 with log-cosh leaves its worst bus slightly *worse* (141 $\rightarrow$ 160). Both end up in the same place: calibrated, G3-mse reaches $R_\sigma^2 = +0.438$ against G3’s +0.630. MSE is not a useful substitute, but its damage is more of the kind a post-hoc offset can undo.

Three conclusions, and the first two overturn statements made earlier in this report.

The models are not failing to track the scenario. After the static profile is removed they still explain 59–75% of the variance of $|V|$. What they get wrong is each bus’s mean level, by more than the spread they are predicting, and that miscalibration is what every negative R^2 in this report and in the four-model report is dominated by. It is also free to remove: no retraining, no architecture change, a 2–2.7 $\times$ reduction in $|V|$ RMSE and 3–4 $\times$ in median $|\Delta Q|$.

The angle advantage is real but roughly a third of its apparent size, and for GridSFM it nearly vanishes: 2.144 $^\circ$ per bus against 0.447 $^\circ$ per edge. Comparisons on per-bus θ RMSE across models with different gauge behaviour are not meaningful.

Neither correction touches the failure in Section 6. The worst-bus $|\Delta Q|$ is unchanged or worse after debiasing, and the constant predictor still beats every debiased model on median residual by an order of magnitude. The stiffness-localized tail on the four 22-kV buses is a separate defect from the calibration one, and only the calibration defect has a free fix.

9 Interpretation and bottleneck

The GridSFM architecture offers plausible explanations for its large-grid feasibility advantage. GridSFM uses seven heterogeneous node types (bus, generator proxy, load, shunt, line, transformer and cycle), promotes branches to nodes, includes a fundamental cycle basis, and performs global linear attention within each node type. It also supplies DC/Laplacian features, effective-resistance landmarks, branch loading features and explicit bus-type encodings. Its angle head is a bounded residual around a DC prior and its magnitude head is clipped to voltage limits. However, the ppNR-v2 run in the four-model report did *not* use the released native GridSFM multi-term objective. Its dispatch used voltage MSE plus a shared PPC masked log-cosh residual loss with physics weight 10^{-2} . Native GridSFM loss components are therefore design references, not an explanation directly tested by that run.

The current PIGNN G0 backbone instead performs repeated learned corrections over a bus-centric one-hop graph. Eight local layers provide a large formal receptive field, but long-range information must still traverse local nonlinearities. The G2/G3 sidecars add one graph token; they do not reproduce GridSFM’s node-level same-type global attention. PIGNN also composes update errors over 40 steps, so a correction that improves voltage MSE can increase the AC residual. This is the principal bottleneck, rather than simply an insufficient value of K .

The breakdown in Section 6 narrows this list considerably. Since PIGNN already matches or beats GridSFM in the bulk and on every $|Y_{ii}|$ bin below 10^4 , the ingredients that would explain a broad representational advantage are not the ones being exercised. What separates the two models is confined to the worst-conditioned 0.2% of buses, and GridSFM’s uniform absence of any entry above 10 pu points at its output bounds. The two candidates worth testing are therefore *bounded terminal corrections* — an output clip, as distinct from PIGNN’s Armijo step bound — and *diagonal equilibration of the residual features*, which targets the $|Y_{ii}|$ dependence directly. Same-type global communication and explicit branch/cycle states remain plausible but are not supported by this evidence: the global-context sidecars G1–G5 span a 2 $\times$ range in tail mass without moving the bulk at all. A two-hop block is a useful intermediate test because it provides a sparse mesoscopic shortcut, but it should be evaluated against a parameter-matched one-hop control and then combined with RANGE- $M = 1$ only if the standalone two-hop result is positive.

10 Metric and campaign caveats

The historical field names `dPinf/dQinf` are potentially misleading. In both the four-model driver and the current PIGNN driver, the per-scenario bus maximum is accumulated and

divided by the scenario count:

$$\text{case-max mean}(P) = \frac{1}{N_s} \sum_{s=1}^{N_s} \max_{i \in \mathcal{M}_P} |\Delta P_{s,i}|.$$

The true global maximum $\max_{s,i} |\Delta P_{s,i}|$ was not reported by the four-model tables; it is now measured for GBnetwork in Section 6, and it separates the models far more sharply than the mean-of-maxima reducer does (G4 reaches 304 pu on P where its mean per-scenario maximum is only 7.3). Pooled mean, median and quantiles instead reduce the full set of valid bus-scenario entries; they must not be called case maxima. A strict head-to-head claim still requires the same split, masks and precision, but not because the historical rows used a different maximum reducer.

The GraphKit/GridSFM campaign also predates the later PIGNN signed-log, global-context and RANGE experiments. Its results should therefore be used as architectural evidence and motivation, not as a strict controlled head-to-head estimate of the latest PIGNN gap.

11 Recommended next step

The immediate decision path is:

1. RANGE- $M = 1, 4, 8$ is finished and test-rescored; treat it as closed and do not extend it to further master counts;
2. compare the two-hop H0–H5 campaign using both mean scenario maxima and the true global P/Q maxima, which Section 6 shows are the discriminating statistic;
3. retain G2 as the voltage-oriented global candidate and G3 as the feasibility-oriented candidate;
4. test the two mechanisms that Section 6 actually implicates — a bounded output head, and diagonal ($|Y_{ii}|$ -equilibrated) residual features — on the four 22-kV buses before spending further effort on global-context topology;
5. if H2 or H5 is positive, run three seeds on the selected GBnetwork candidate before extending to additional grids.